



Artificial Intelligence in Aviation Safety: Systematic Review and Biometric Analysis

Gülay Demir¹ · Sarbast Moslem^{2,5} · Szabolcs Duleba^{3,4}

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Abstract

This study aims to offer aviation safety researchers, practitioners, and decision-makers a comprehensive exploration of integrating advanced technologies, such as artificial intelligence and machine learning, to inform and fortify future safety strategies. Focusing on systematic and bibliometric perspectives, the paper reviewed 224 articles in the Scopus database from 2004 to 2024 (January). Key findings highlight China's notable contributions to aviation safety research, underscoring its leadership in international collaboration. The techniques employed encompass machine learning, time series models, deep learning, AI, neurophysiological modeling, and optimization algorithms. The analysis discerns prominent research trends, including aviation accident analysis, pilot behavior, aviation safety measures, and endeavors to enhance safety standards. The aviation industry's steadfast commitment to safety, efficiency, and technological innovation is evident. By uncovering the main structures, foci, and trends in aviation safety research, this study equips researchers and practitioners with crucial insights into ongoing endeavors and potential future developments, fostering a more profound understanding of aviation safety.

Keywords Aviation safety · Artificial intelligence · Biblioshiny · VOSviewer · Systematic review · Bibliometric analysis

1 Introduction

Traditionally, aviation safety research has been conducted by analyzing accident and incident reports, pilot training, and improving procedures in aviation operations. However, in recent years, with advanced technologies, such as artificial intelligence (AI), machine learning (ML), and deep learning (DL), playing an important role in aviation safety, new methods of analysis and analysis have emerged [9]. These technologies offer significant advantages in analyzing large

amounts of data, recognizing patterns, and identifying potential safety risks [112].

Aviation safety is a top priority in the industry, where the well-being of passengers and crew and the efficient functioning of air travel are paramount [65]. As the industry evolves, embracing technological advancements like AI becomes crucial. AI transforms traditional safety measures by introducing predictive analytics, real-time monitoring, and proactive risk management. Through machine learning (ML) algorithms, a subset of AI, vast datasets are analyzed to identify patterns and trends, enabling the prediction of potential safety issues before they escalate. Furthermore, AI facilitates the development of advanced decision support systems that assist pilots and air traffic controllers in making informed and timely decisions [96]. By leveraging AI-powered predictive maintenance, airlines can identify and address potential mechanical issues before they compromise safety, assessing various factors, such as aircraft performance data and maintenance records, to predict when components may require attention, thus reducing the likelihood of in-flight failures [93–95, 131].

Additionally, AI enhances air traffic management systems by optimizing routes and improving communication between aircraft and control towers, which not only boosts efficiency

✉ Szabolcs Duleba
duleba.szabolcs@kjk.bme.hu

¹ Sivas Cumhuriyet University, Sivas, Turkey

² School of Architecture Planning and Environmental Policy, University College Dublin, Dublin, Ireland

³ Budapest University of Technology and Economics, Budapest, Hungary

⁴ University of Nyíregyháza, Nyíregyháza, Hungary

⁵ Research Center of Digital Economics, Azerbaijan State University of Economics, UNEC, Istiglaliyyat Str. 6, Baku, Azerbaijan

but also contributes to the overall safety of air travel [41]. This comprehensive study delves into the intricate landscape of aviation safety research, employing a systematic and bibliometric approach to explore the multifaceted applications of AI, ML, and deep learning (DL) techniques. Pursuing heightened aviation safety extends beyond regulatory compliance, representing a fundamental endeavor to elevate safety standards and fortify air transport systems.

AI, ML, and DL offer significant advantages over traditional probability theory and mathematical statistics-based methods in managing safety performance in aviation. First, AI's ability to simulate human-like intelligence greatly enhances decision-making processes in aviation safety. While traditional methods often rely on fixed analyses based on specific data, AI can analyze a wide range of data and dynamically assess situations. This flexibility allows for the development of rapid and practical solutions in unexpected scenarios. ML's capacity to identify patterns within large datasets enables a proactive approach to aviation safety. Traditional probability theory typically makes predictions based on historical data, but ML algorithms can extract meaningful patterns from vast and complex datasets. This capability allows for the early detection of potential hazards and interventions before these hazards manifest.

Additionally, ML continually learns and updates itself, contributing to improving safety performance. DL's superiority in handling complex information hierarchies extends beyond traditional statistical methods. DL can quickly process large volumes of data and uncover deep relationships that may have been previously overlooked. This is particularly valuable in evaluating safety risks within the highly multidimensional and complex systems in aviation. For example, DL can analyze multiple variables during a flight in real time and predict potential risks based on this data, providing a significant advantage in risk assessment. In this study, the advantages provided by AI, ML, and DL over traditional methods represent a transformative shift in managing aviation safety. These technologies offer predictive analytics to foresee potential issues and enable rapid responses through real-time monitoring systems. As a result, integrating AI, ML, and DL into aviation safety plays a critical role in enhancing the resilience and reliability of air transport systems, significantly strengthening global aviation safety.

One of the significant contributions of this study to the literature and practice is its systematic and bibliometric approach to examining research in aviation safety, highlighting the multifaceted applications of AI, ML, and DL techniques. The unique contributions of this study to the literature and practice are particularly prominent in its in-depth examination of the impacts of AI, ML, and DL techniques on aviation safety. The bibliometric analysis method used in this study allows researchers to understand this field's current state better and identify potential areas for future research.

This study underscores the importance of integrating AI, ML, and DL techniques to enhance aviation safety standards and strengthen air transportation systems, thoroughly examining the valuable contributions made by research in this area to both academic literature and practical applications.

The research questions (RQ) aim to systematically and bibliometrically assess the impact and development of AI, ML, and DL techniques on aviation safety in the literature.

- RQ1: What are the trends in using AI, ML, and DL techniques in aviation safety literature over time?
- RQ2: Can it be determined which sub-fields AI, ML, and DL research on aviation safety is focused on and which topics are prominent?
- RQ3: Which countries or organizations are the most researched and published groups on AI, ML, and DL techniques in aviation safety?
- RQ4: Can it be analyzed around which keywords the publications in the literature are concentrated on and which sub-fields these words represent?
- RQ5: In which journals are AI, ML, and DL research in aviation safety concentrated, and how do these publications interact?

Consequently, this paper is a guiding resource for future research exploring the use of AI, ML, and DL techniques in aviation safety.

This paper is organized as follows: Sect. 2 overviews the methodology. Section 3 conducts a systematic review of aviation safety research; Sect. 4 presents a bibliometric analysis of AI in aviation safety; and the final two sections summarize the main findings and emphasize the study's limitations.

1.1 Background of the Research

This study is distinguished by its in-depth examination of the impacts of AI, ML, and DL techniques on aviation safety. One of the significant contributions of this study to the literature and practice is its systematic and bibliometric approach to examining research in aviation safety, highlighting the multifaceted applications of AI, ML, and DL techniques. The unique contributions of this study to the literature and practice are particularly prominent in its in-depth examination of the impacts of AI, ML, and DL techniques on aviation safety. The bibliometric analysis method used in this study allows researchers to understand this field's current state better and identify potential areas for future research. This study underscores the importance of integrating AI, ML, and DL techniques to enhance aviation safety standards and strengthen air transportation systems, thoroughly examining the valuable contributions made by research in this area to both academic literature and practical applications. In essence, this research contributes not only to the

academic discourse but also to the practical realm of aviation safety, fostering a deeper understanding of the transformative impact of AI, ML, and DL in fortifying the resilience and reliability of air transport systems worldwide.

2 Review Methodology

Bibliometric analysis is a powerful technique for tracking research trends and discoveries. The initial step involves selecting an appropriate database. Accessible data sources include Scopus, Web of Science (WoS), MEDLINE, Cochrane Library, PubMed, Science Direct, IEEE Xplore, EBSCO, Taylor & Francis, and Springer. After selecting the database, the information is filtered by applying “inclusion” and “exclusion” criteria. One or more software programs, such as VOSviewer, CiteSpace, SciMAT, BibExcel, Publish or Perish, HistCite, Biblioshiny (R Package), Sci2 Tool, CitNetExplorer, and BiblioMaps, can be used together to perform bibliometric analysis.

2.1 Search Strategy

A comprehensive search was conducted in the Scopus database, focusing on literature related to “aviation safety” and associated keywords, such as “air safety”, “flight safety”, “runway safety”, “air traffic safety”, “air traffic control safety”, “cabin safety”, “helicopter safety”, “taxiway safety”, “airport surface safety”, “aviation accident”, “aviation incident”, “aviation mishaps”, “aviation safety management”, “aviation risk”, “risk in aviation”, “aviation hazards”, “human factor in aviation”, “pilot situational awareness”, “human aviation factor”, “human error in aviation”, “pilot fatigue”, “aviation safety performance”, “safety occurrences in aviation”, “aircraft icing”, “bird strike in aviation”, “wild-life strikes in aviation”, “loss of control in aviation”, “pilot error”, “maintenance error in aviation”, “aviation resiliency”, “aircraft accident investigation”, “aircraft crash”, “aircraft turbulence”, “safety of unmanned aerial vehicles”, “safety regulation in aviation”, “cockpit and safety”, and “machine learning”, or AI. The publication period was limited to January 2024, resulting in 224 relevant articles.

2.2 Conducting Phase

Relevant literature on aviation safety published in different academic journals was included. The articles were written in English and categorized as “articles”. Conference abstracts, letters, theses, editorials, book reviews, biographies, publications withdrawn by authors, news reports, conference presentations, dissertations, and duplicate publications are excluded. Only research articles published in journals

have been included. This is because these studies have been reviewed by other scientific experts working in the same field and provide more rigorous assurance than other documents. Conference papers have been excluded as they are not peer-reviewed [122, 184].

2.3 Selection Criteria

In the current study, SciVerse Scopus was selected to achieve the objectives. Scopus is a bibliographic database created by Elsevier in November 2004 (Elsevier; Scopus, 2024). With over 94 million records, more than 29,200 active serial titles, and over 330,000 books, Scopus offers the most comprehensive view of the world’s research output in the fields of science, technology, medicine, social sciences, arts, and humanities (Elsevier; Scopus, 2024). The Scopus database includes all MEDLINE journals and allows for citation analysis [48, 85]. The Scopus database has various operational functions that facilitate bibliometric analysis. Among these operational functions are the journal name, document type, publication year, authors and their affiliations, citation count, and h-index metrics for documents [2]. Even though WoS and Scopus databases have a high association, i.e., overlap in journal indexing, they also index different journals [20, 115, 163]. Thus, using Scopus is an attempt to cover more topics that may not be available in WoS and may not have been discussed [119, 166]. Based on inclusion and exclusion criteria, the literature screening process resulted in 224 articles. The PRISMA flow diagram (Fig. 1) illustrates the screening process.

The screening involved three stages: identification, screening, and inclusion. Initially, 261 records were identified. After screening, 37 records were excluded, resulting in 224 articles in the final review.

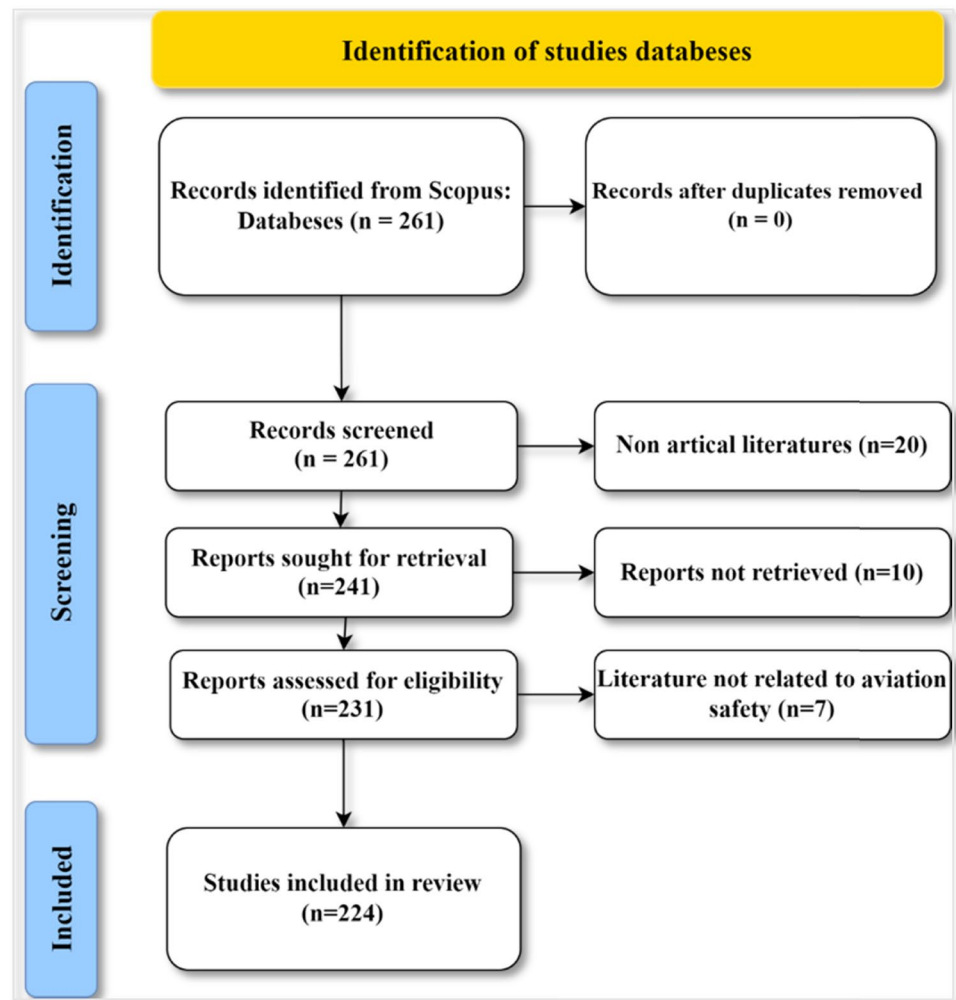
2.4 Bibliometric Analysis Software

The strengths and the weaknesses of the commonly used bibliometric analysis software VOSviewer, CiteSpace, SciMAT, BibExcel, Publish or Perish, HistCite, Biblioshiny (R Package), Sci2 Tool, CitNetExplorer, BiblioMaps are summarized in Table 1.

Biblioshiny was used in the study because it is a user-friendly, internet-based application that does not require coding [10]. In addition, VOSviewer was used to create network mapping [162].

2.5 Classification of Bibliometric Analysis

To systematically examine aviation safety research, various classification methods were utilized:

Fig. 1 Data collection PRISMA flow diagram

- **Bibliographic Coupling:** This method links documents referencing the same sources, indicating related topics within aviation safety research. It helps identify the existing research network and related studies.
- **Co-Citation Analysis:** This method evaluates how frequently specific articles are co-cited in other studies, identifying critical articles central to aviation safety research.
- **Keyword Analysis:** The study assessed the popularity and trends of particular topics within aviation safety research by examining the keywords used in the articles.
- **Time Series Analysis:** This analysis looked at the number and topics of articles published over time, revealing trends in aviation safety research from 2004 to 2024.

Using these methods, the study provided a clearer picture of the existing knowledge network and potential future research areas in aviation safety.

3 Systematic Review of AI in Aviation Safety

The systematic review was used to identify the main structures and focal points of research on aviation safety. The systematic analysis revealed which sub-topics are emphasized, which techniques are used more frequently, and research trends. The distribution of the 224 studies analyzed according to their primary areas of interest is given in Fig. 2.

Each study in Fig. 2 has a unique focus in areas, such as aviation safety and data analytics, and has addressed specific issues in the aviation sector using different techniques and methodologies.

The studies were categorized into thematic categories and classified as follows:

- **Accident Prevention and Prediction**
 - **Accident Prediction and Analysis:** Ziegler et al. [204], a mathematical model for probabilistic pre-

Table 1 Bibliometric Software and Their Strengths and Weaknesses. Source: <https://www.vosviewer.com/>, <https://citespace.podia.com/>, <https://sci2s.ugr.es/scimat/description.html>, <https://homepage.univie.ac.at/juan.gorraiz/bibexcel/>, <https://publish-or-perish.en.softonic.com/>, <https://histcite.software.informer.com/>, <https://www.bibliometrix.org/home/index.php/layout/biblioshiny>, <https://sci2.cns.iu.edu/user/index.php>, <https://www.citnetexplorer.nl/>, <https://bibliomaps.es/>

Software	Strengths	Weaknesses
VOSviewer	Excellent visualization capabilities for bibliometric networks User-friendly interface Free and open-source	Limited advanced statistical analysis features Issues with large datasets may require preprocessing
CiteSpace	Versatile with many functions for trend analysis Strong in temporal analysis (e.g., research fronts, burst detection) Free	Steep learning curve; not beginner-friendly Java dependency can complicate installation
SciMAT	Great for longitudinal analysis of scientific topics Strong science mapping tools	Complex interface, challenging for beginners Limited export formats for data and visualizations
BibExcel	Highly flexible, customizable analysis Efficient at processing large datasets Free	Lacks built-in visualization tools A steep learning curve with minimal guidance is available
Publish or Perish	Excellent for citation analysis using Google Scholar Simple and user-friendly Free	Data may need to be more accurate due to reliance on Google Scholar Lacks advanced features found in other tools
HistCite	Strong in visualizing citation histories and relationships Simple and user-friendly	Development is discontinued; it may not work with new datasets Mainly works with Web of Science data, limiting its application
Biblioshiny (R Package)	Comprehensive analysis features Integration with R allows advanced statistical analysis Free and open-source	Requires familiarity with R, which may be a barrier Visualization capabilities are less robust than those of VOSviewer
Sci2 Tool	Excellent for comprehensive scholarly data analysis Strong support for temporal, geospatial, topical, and network analysis Free	Difficult for beginners due to steep learning curve Interface needs to be updated compared to other tools
CitNetExplorer	Specializes in citation network analysis Allows interactive exploration of citation networks	Primarily focused on citation networks, lacking broader functionalities Requires time to master, especially for users new to network analysis
BiblioMaps	Strong mapping capabilities, particularly geographic distributions Allows for custom configurations	Niche tool with a narrower focus Limited community support and resources for troubleshooting

diction and explanation of NMACs (Near et al.) for aviation safety and data analysis. Omrani et al. [130] used ML techniques to predict the severity of aviation accidents. Monika et al. [120] used time series models to analyze aviation accidents and predict future accidents. Caetano [18] predicts aircraft accidents and incidents using meteorological data—and Guo et al. [54] predict accident types during approach and landing. Nogueira et al. [127] predict fatal situations contributed by human factors in aviation incidents. Li et al. [93–95], accurate prediction and interpretation of aviation safety events.

- Accident Prevention and Safety Measures: Zhang et al. [197, 198] developed a model for recognizing electrical poles and assuring aviation safety

when flying agricultural uncrewed aerial vehicles. Jiang et al. [71], a real-time semantic segmentation method for identifying emergency landing zones for UAVs. Inan [63], aircraft damage classification and aviation safety enhancement. Wang [164], fast and accurate detection of foreign objects on airport runways. Li et al. [93], detecting damage to aero-engine blades and improving aviation safety. Lin et al. [101, 102], prediction of landing gear performance and a learning model based on a dataset. Liang et al. [97], fast and accurate identification of airport runway degradation. Khattak et al. [75], prediction of aircraft retraction under wind shear conditions. Choi and Gibson [30], a study of the impact of COVID-19 on aircraft safety events. Zhang et al. [197, 198] developed a model

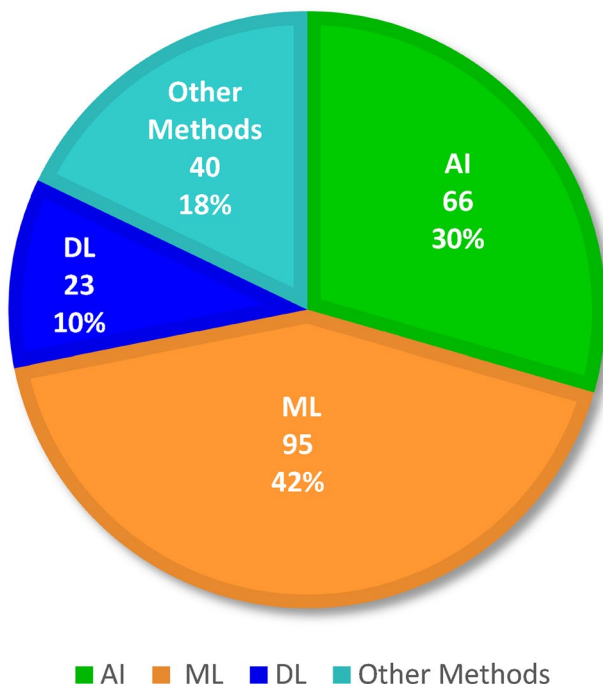


Fig. 2 Distribution of studies on aviation safety by application area

for recognizing electrical poles and assuring aviation safety when flying agricultural uncrewed aerial vehicles.

• Flight Behavior and Pilot Monitoring

- **Pilot Behavior and Condition:** Chen et al. [24–26], scalable deep time-sensitive attention network for explaining hard landing predictions. Alreshidi et al. [7, 8], multimodal approach for EEG (electroencephalography) based detection of pilots' mental states. Alreshidi et al. [9], improving aviation safety with ML and psychophysiological data on pilot behavior. Taheri et al. [158] distinguish the cognitive workload of pilots during flight with electroencephalogram (EEG). Li et al. [93–95] identify pilots' dangerous visual flight rules (VFR) flight behavior. Dong et al. [45], intent modeling of single-pilot operations in commercial aviation. Bi et al. [13], fatigue driving detection technique using Imaging Photoplethysmography (IPPG) technology.
- **Flight Performance and Maneuver Recognition:** Khattak et al. [76], predicting the air path of wind-driven aircraft. Lu et al. [109], DL techniques for flight maneuver recognition. Khattak et al. [79], ML models predict flight performance and improve aviation safety.

• Environmental Factors and Weather Conditions

- **Weather Conditions and Safety:** Chen et al. [24–26], aviation visibility is integrated with convolutional neural network (CNN) and long short-term memory (LSTM) networks. Khattak et al. [77], estimation of Missed Approach (MAP) and its application as a safety-critical process. Khattak et al. [78], severe low-level wind shear on airport runways is classified, predicted, and interpreted. Khattak et al. [81], identification and evaluation of severe wind shear risk factors around airport runways. Yu et al. [186, 187] the vertical development of clouds and their impact on aviation safety. Xu et al. [177, 178] investigated the association between bird activity and weather and forecasted bird activity. Chkeir et al. [29] forecast extreme weather events quickly. Petracca et al. [140], volcanic clouds may be accurately and automatically detected using satellite data. Wang et al. [172–174], training neural networks using physics knowledge to understand 3D clouds. Niu et al. [126] consider heterogeneous data in precipitation prediction.

• Risk Management and Decision Support Systems

- **Risk Management and Decision-Making:** Cankaya et al. [17], evidence-based management decisions: Bayesian inference in aviation events. Liu et al. [108], joint optimization of risk factors and energy consumption in Internet of Things (IoT) networks with Internet of Unmanned Aerial Vehicles (IUAVs). Lin et al. [101], risks and user experience of AI-based techniques to improve aviation safety. Hancock [58], how to react, respond, and train human-machine systems to rare, uncertain, and unexpected events for computing and data analysis.
- **Data Analysis and Anomaly Detection:** Gao et al. [52] constrained LSTM (Long et al.) autocoder for flight data outlier detection. Coelho e Silva and Murca [34], a data mining project for detecting anomalies during aircraft operations. Choi et al. [31], stochastic fitness anomaly detection and resolution for air traffic control.
- **Aircraft Design and Performance Optimization**
 - **Aircraft Design and Innovations:** Divya et al. [41], micro-patch antenna design with AI integration for AMCA aircraft. Qi et al. [144] weakly supervised deep-level set learning for identifying aero-engine defects. Wang and Zhao [165], data-driven forecasts of exhaust gas temperature baselines for aero engines. Kilic et al. [82], ML models are focused on aviation safety, aircraft performance, and engine health for performance analysis and prediction. Guraksin and Ozcan [55], improving aviation maintenance, repair, and overhaul (MRO) planning and integration with

product lifecycle management. Li et al. [93], detecting damage to aero-engine blades and improving aviation safety.

- **Air Traffic Management and Systems**
 - **Air Traffic Management and UAV Systems:** Xu et al. [177, 178], air traffic density estimate using a Bayesian ensemble graph attention network (BEGAN). Danzyuryun and Kalyagin [35], user services and database coordinate system selection of Russian unmanned aircraft system traffic management (RUTM), Russia's UAV (Uncrewed et al.) traffic control system. Yang et al. [182, 183], detection of abnormal situations to ensure the aviation safety of UAVs. Yu et al. [186, 187], improving balanced motion repetition of quadrotors with deep reinforcement learning. Liu et al. [107], for UAVs and security, the discovery of physical hostile attacks against UAVs and taking countermeasures. Sai et al. [151] analyze the current and future developments in the usage of AI in UAVs. Rahmani and Weckman [146] analyze occupational safety risks commercial drone pilots face for operational procedures and techniques.
- **Safety and Precautions**
 - **Safety Measures and Monitoring:** Schekler et al. [153], automatic detection of migratory bird flocks using weather radars and DL. Han et al. [57], urban air transport and how new aircraft can affect the air communication spectrum: Yang et al. [182, 183], anomaly detection and explanation for the safety and reliability of UAVs.
- **Technological Developments and Applications**
 - **Data Mining and Innovative Analysis Methods:** Ziakas and Pechlivanis [203], AI applications for classifying aviation accidents. Wang et al. [172–174], data mining and innovative analysis methods are used to solve problems related to aviation safety and performance.

These categories highlight the specific research focus of each reference, emphasizing the broad spectrum of topics, such as aviation safety, pilot performance, environmental impacts, risk management, air traffic management, aircraft design, and technological innovations.

3.1 Methods Used in Analyses

The techniques used in aviation security studies can be classified into various categories based on their methodologies and applications. These include statistical methods, machine learning algorithms, deep learning models, signal processing techniques, optimization methods, and hybrid approaches. Each category enhances aviation safety, accident prevention, and risk management. Below is a detailed classification and explanation of these techniques:

- **Statistical and Probabilistic Methods**

Statistical techniques, such as Bayesian networks and Bayesian optimization, have been widely used in aviation safety studies. Bayesian networks [17, 204] provide a probabilistic graphical model that represents a set of variables and their conditional dependencies via a directed acyclic graph. Bayesian optimization [76] is another technique that helps optimize complex functions with fewer evaluations, making it highly suitable for risk management in aviation—multidimensional interpolation [189].

Other statistical methods include time series models like autoregressive integrated moving average (ARIMA), which are used to predict future events based on historical data [120]. Logistic regression and discriminant analysis are traditional statistical approaches for classifying events and understanding variables' relationships [65].

- **ML Algorithms**

ML techniques are extensively applied in aviation security for prediction, classification, and anomaly detection. Commonly used algorithms include decision trees, support vector machines (SVM), and random forest models [177, 178]. Decision trees and SVMs [130] are utilized for classification tasks, such as predicting the severity of aviation accidents. Artificial Neural Network (ANN) includes Back Propagation Neural Network (BPNN), Radial Basis Function Neural Network (RBFNN), and Apple Neural Network [54]. Text-Based ML Techniques [148]. Extreme Learning Machine (ELM) [168–171]. Advanced ML models like XGBoost [44] and ensemble methods enhance the accuracy of predictions. Random forest models are particularly effective in handling large datasets and providing robust classifications. Additionally, ML models are frequently combined with data preprocessing techniques to improve performance [93–95, 113, 200].

- **DL Models**

DL techniques are increasingly adopted in aviation for their ability to model complex patterns in large datasets. Convolutional Neural Networks (CNNs) and their variants are widely used for image and signal processing tasks. For instance, MSCNN (Multiple et al.) [114, 153, 182, 183] is used for safety and reliability analysis in UAVs, while deep CNN [92, 96, 167] models are applied to detect runway degradation. Convolutional-LSTM (ConvLSTM) and Trajectory-Gated-Recurrent-Unit (TrajGRU) [186, 187]. LSTM Encoder-Decoder (LSTM E/D) [29]. Recurrent Neural Networks (RNNs) and their derivatives, such as LSTM (Long et al.) and Gated Recurrent Unit (GRU), are employed for sequential data analysis. LSTM and bidirectional LSTM (Bi-LSTM) models [120] are particularly effective for time series prediction, and

LSTM autoencoders [52] are utilized for anomaly detection in-flight data.

- *Signal Processing and Physiological Data Analysis*

Signal processing techniques are crucial in monitoring and analyzing the physiological states of pilots and other crew members. Techniques like Imaging Photoplethysmography (IPPG) [13] and electroencephalography (EEG) are used to detect fatigue and cognitive workload in pilots. Riemannian geometry analysis [7, 8] and multimodal approaches [7, 8] further enhance the accuracy of detecting pilots' mental states by analyzing complex physiological signals.

- *Optimization Methods*

Optimization techniques are essential for improving the efficiency and safety of aviation systems. Methods, such as Monte Carlo simulations [59], ant colony optimization [175], and the Whale Optimization Algorithm-based Artificial Neural Network (WOANN) technique [11], are used for optimizing various aspects of aviation operations, from flight paths to resource allocation. Crow Search Algorithm (CSA) [168–171]. Kernel-based methods like Least Squares Support Vector Machines (LS-SVM) [22, 23] and Kernel principal component analysis [201] are also employed for dimensionality reduction and feature selection, which are crucial for handling high-dimensional data in aviation studies.

- *Hybrid and Advanced Techniques*

Several hybrid techniques combine multiple methodologies to improve performance and accuracy. For instance, the multi-agent deep reinforcement learning VD3QN algorithm [57] integrates reinforcement learning with DL to optimize UAV navigation and control.

Similarly, the MHCNN-RNN (Multi-Head et al.) [89] combines CNNs with RNNs for comprehensive sequence and spatial analysis. AI techniques, including ChatGPT [203], are also being explored for natural language processing tasks in aviation safety, such as Named Entity Recognition (NER) and computerized language processing (CLP). These approaches are instrumental in analyzing text data related to aviation incidents and extracting relevant information for safety analysis.

- *Innovative Modeling Techniques*

Innovative modeling techniques like the physical inverse learning machine (FP_ILM) algorithm [49] and the Multiple Correlation Analysis (MCA) models [101, 102] are used for specialized tasks in aviation safety, such as identifying aero-engine defects and predicting landing gear performance. Additionally, AI integration into antenna design [41] and deep-level set learning [144] further exemplify the application of advanced modeling techniques in aviation—fuzzy-Logic Modeling [168–171].

Many of these techniques are used in aviation safety and data analysis. For example, ML algorithms, time series models, and DL techniques are widely used to analyze aviation data and improve aviation safety. Furthermore, advanced technologies, such as AI techniques and neurophysiological modeling, are essential in analyzing pilot behavior and improving aviation safety. These are crucial to the development of data analysis and aviation safety initiatives.

The aviation industry is constantly focused on research and innovation related to safety, efficiency, and technological

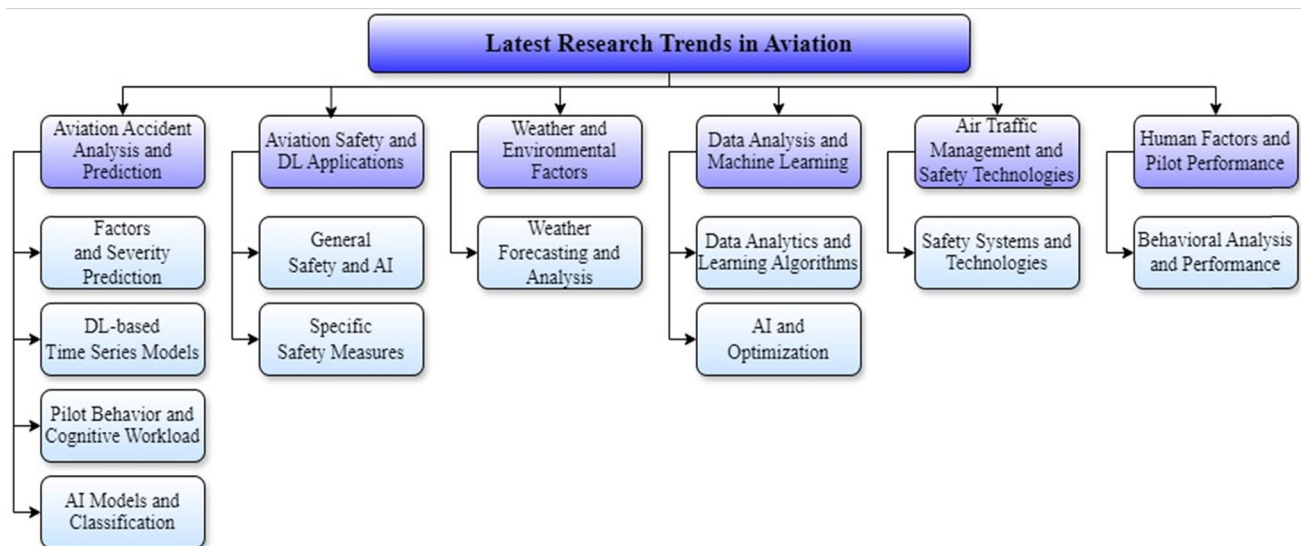


Fig. 3 Latest research trends in aviation

developments. Figure 3 shows the latest research trends and academic studies in the field of aviation, with a focus on examining critical issues according to the main themes.

This classification organizes the related studies according to the main themes, allowing each study to be examined in a specific context.

- *Aviation Accident Analysis and Prediction*

- **Factors and Severity Prediction:** Determining relationships between various contributing factors in aviation accidents and developing severity prediction models [130]. Accident factors, survival rate analysis, and ML models [65]. Aircraft accidents, incident detection, and data analytics [181].
- **DL-based Time Series Models:** Investigating the usefulness of DL-based time series models for the study and prediction of accidents [120].
- **Pilot Behavior and Cognitive Workload:** Conducting more comprehensive research on pilot behavior and improving model performance [9]. Analyzing the cognitive workload of pilots with EEG data and ML methods and contributing to the development of flight systems [158]. Monitoring pilots' emotional states and effectiveness of flight training [172–174]. Mental workload classification, analysis of brain activity, pilot performance [197, 198].
- **AI Models and Classification:** Developing AI models for analyzing and classifying aviation accidents [203].

- *Aviation Safety and DL Applications*

- **General Safety and AI:** Aviation safety, analysis of EEG signals, AI-supported safety systems [93–95]. Aviation safety, DL applications, and time series data analysis [52]. Aviation safety, model explainability, causes of hard landings [24–26]. Runway safety, object recognition, and DL applications [164]. Air traffic management, complex system models, DL applications [177, 178]. Turbulence analysis, DL techniques, and data collection [150]. Air traffic safety, AI applications, and modeling [131]. Aeroicing analysis reduced ranking models and real-time data analysis [189].
- **Specific Safety Measures:** Wing damage detection, DL methods, and image processing [93–95]. Icing modeling, ML algorithms, aviation safety [135]. Engine fault detection, data analytics, and learning algorithms [202]. Aircraft carrier landing optimization, ant colony algorithm, safety analysis [175]. Accurate estimation of cloud base height for aviation safety [159]. Identification and prediction of hazards

for civil aviation safety [199]. Aircraft icing and actuator/sensor failure detection [42].

- *Weather and Environmental Factors*

- **Weather Forecasting and Analysis:** Using a neural network ML approach to classify turbulence intensity based on radar observations [19]. ML detects unusual weather conditions, and automatic dependent monitoring-broadcast (ADS-B) data is used [67]. Weather forecasting, wind shear detection, and artificial neural networks [88].

- *Data Analysis and ML*

- **Data Analytics and Learning Algorithms:** Categorization of event risk levels, development of an ML model, extension of risk level prediction to event level results [190]. Prediction of engine failures and assessment of engine health by ML [111]. Analysis of EEG signals, DL models, and fatigue recognition [56]. Flight data records, cluster analysis, supervised and unsupervised learning [33]. Classification of pilot intent, classification methods in complex environments [168–171].
- **AI and Optimization:** Cognitive systems, human-machine interaction, aircraft operations [103]. Aircraft utilization optimization, AI applications, dynamic time compression, and genetic programming [149]. Fuel consumption prediction, air emissions, and flight data analytics [5].

- *Air Traffic Management and Safety Technologies*

- **Safety Systems and Technologies:** Air traffic management, aviation safety, ADS-B/TIS-B technologies [132]. Obstacle detection, aircraft safety, and aviation safety [50].

- *Human Factors and Pilot Performance*

- **Behavioral Analysis and Performance:** Driver behavior, attention deficit, transport system optimization [128].

3.2 Aviation Safety Use of AI

This chapter reviews how AI, ML, and DL techniques are used in aviation safety. These techniques can contribute to aviation safety in many areas, such as data analysis, prediction models, pattern recognition, and automated decision-making.

Various research studies in aviation focus on various topics, from safety to efficiency to technological advances. The papers presented represent research in various disciplines that address essential issues in the aviation industry and include contemporary technologies such as AI.

- **Mathematical and Statistical Modeling Techniques:** Ziegler Haselein et al. [204] this study presents a mathematical model for predicting Near-Mid-Air Collisions (NMACs) based on the NASA ASRS (Aviation et al.) database. Jiang et al. [70], this research uses visual gaze and operational behavior analysis techniques to determine the intelligence level of pilots. Jiang et al. [69], the study develops a diagnostic framework for cause-and-effect reasoning in aviation safety accidents using non-axiomatic logic. Yan et al. [180] developed a virtual airplane maintenance training system based on gesture recognition and the PSO-BPNN (Particle et al.) model.
- **Artificial Intelligence and Machine Learning (AI/ML):** Ziakkas and Pechlivanis [203], this study explores the application of AI in the classification of aviation accidents. Shafi et al. [154] use an artificial neural network (ANN) model for spare parts prediction and classification, impacting aviation safety. Wang et al. [172–174], multi-modality fusion is used in emotion recognition methods during pilot training. Lin et al. [101, 102], this research investigates the technical dangers of AI-enabled procedures for aviation safety monitoring in voice communication. De Giorgi et al. [36], engine health is monitored by combining an artificial neural network (ANN) with kernel-based principal component analysis (PCA). Massé et al. [116], the classification of electrophysiological signatures in a flight simulator is performed using explainable AI. Wang et al. [168–171], time-dependent pilot intention determination is conducted using the ESAX-CSA-ELM classification approach in complex situations. Chen et al. [22, 23] use the LS-SVM algorithm to improve high plateau flight data matching. Jing et al. [72] present a random forest-based approach for detecting foreign body residues using optical image sensors. Ortnier et al. [131], an AI-enabled digital assistance system developed for forecasting air traffic conflicts. Dong et al. [45], AI technology is applied in modeling single-pilot intention in commercial aviation. Liu et al. [104], air traffic control data is exploited using Received Signal Strength Indicator (RSSI) and Time Stamps at Server (TSS) for GPS (Global et al.) spoofing detection. Ni et al. [125] an ensemble classification model is developed for aviation safety using unbalanced data. Zhang and Mahadevan [191], Bayesian neural networks are used for flight route prediction and safety evaluation. Blanco et al. [15], this study measures cognitive workload using passive, dry EEG measurements in simulated flights.
- **Safety and Operational Efficiency:** Prosvirina and Tikhonov [142], aviation safety is improved by creating uncrewed aerial vehicles domestically, replacing imports. Rahmani and Weckman [146], the study investigates occupational safety risks for commercial drone pilots. Saleh et al. [152], this research provides new data-driven insights and analysis on tail rotor failure in helicopter crashes. Pérez-Castán et al. [138], learning assurance analysis is conducted for designing and developing dependable AI-based systems in aviation. Holford [60] an ethical investigation is conducted on the effects of cockpit automation on airline pilots' responsibilities. Abin et al. [1] propose a framework for using social media in-flight route safety assessment. Smirnov [155] explores the use of supercomputers and AI to ensure the safety of space flights.
- **Technological Innovations:** Danzyuryun and Kalyagin [35], user interface requirements are formulated for traffic control systems for uncrewed aerial vehicles in Russia. Liu et al. [107] AI-enabled physical attacks that can mislead perceptions of uncrewed aerial aircraft are investigated. Wang et al. [172–174] comprehensive research is conducted on the use of AI for uncrewed aerial vehicles. Pan et al. [134] developed a method for power line detection using spatial context variation to improve the ability of aircraft to detect small obstacles at low altitudes. Lin et al. [99] designed an operational boundary diagram for a ship-based helicopter using AI techniques. Jiang et al. [70] developed a method to determine the intelligence level of pilots using visual gaze and operational behavior data. Bi et al. [13] proposed a method based on IPPG that proposes detecting fatigue by analyzing driver face changes.
- **Aviation Safety Data Analysis:** Ul Abedin et al. [161], a weakly supervised semantic dictionary, is built to identify causes in aviation safety incident reports. Chouraqui and Doniat [32], the S-ETHOS (System- in French (Etude de l'Homme en Situation de pilotage)) system uses a knowledge-based system for systematic flight analysis, emphasizing human aspects. Emanuilov and Dheu [47], the European Aviation Safety Agency's AI roadmap, is analyzed and evaluated.

ML is a branch of AI that enables computer systems to discover patterns and relationships through data analysis. This technology allows algorithms to learn from data-driven experiences to fulfill a specific task. ML is essential in the aviation industry in many areas, such as aviation safety, efficiency, maintenance predictions, and traffic management. Collecting and analyzing large amounts of data helps improve decision-making processes in the aviation industry and increase operational efficiency, and investigates the evaluation of ML techniques for the severity prediction of aviation accident datasets. The categorization encompasses a wide range of studies using different techniques to address various challenges in aviation, including safety, operational efficiency, and technology integration.

- **Intelligent Temporal and Boundary-Sensitive Detection:** Hu et al. [61], this study focuses on an intelligent temporal detection network that works in a boundary-sensitive manner in the flight regime recognition process, with specific model structures designed for flight data.
 - **ML and AI in Aviation Safety:** Borjalilu et al. [16] integrated ML multi-class classification models, and Markov chains were used to predict cockpit crew performance. Su et al. [156] proposed an improved gray correlation analysis and a combined prediction model to identify and predict aviation accidents. Wang et al. [168–171], a method for detecting obstacles by UAVs in agricultural land, aims to provide solutions to aviation safety problems. Jiao et al. [68], ML is applied to categorize and determine the reasons for Chinese civil aviation incident reports. Zhang et al. [193–196] use a spatio-temporal graph convolutional neural network for airport surface motion prediction and safety evaluation. Liu et al. [105, 106], ML-based non-mesh surface wind speed field reconstruction. Chen et al. [22, 23], detection of abnormal driving behavior of cockpit pilots with You Only Look Once version 4 (YOLOv4) and attention mechanism. Lin et al. [100], daytime monolayer cloud base height is estimated using improved base image data. Kaakai [73] provide a phase in the ML development lifecycle for product certification and approval in aviation. Cheng et al. [27], ML-assisted route estimation and conflict detection for the internet of aircraft. Zhao et al. [201], RIS (Reconfigurable et al.)-enabled air-to-ground NOMA (Non-Orthogonal et al.) communication with a distribution-robust DRL (deep reinforcement learning) approach. Inan and Inan [64], ML approach for analyzing key factors influencing the deadliest accidents in aviation history. Odisho et al. [129], ML application to improve runway safety by reducing runaway risk. Patriarca et al. [137], democratizing business analytics and ML for air traffic management safety. Wang et al. [168–171], transfer learning using extreme learning machines for air engine trouble diagnostics, flight test sensor problem diagnosis using data fusion and ML methods. Zhang et al. [193–196], LiDAR (Laser et al.)-based wind shear detection using statistical characteristics. Inan [62], classification of surviving/non-surviving passengers in deadly aviation incidents using ML. Isufa et al. [66], multiple-drone conflict resolution via graphical convolutional reinforcement learning. Kim et al. [83], ML-based data-driven strategy for real-time flight route optimization. Qin et al. [145] detected mental tiredness using heart rate variability and eye metrics during simulated flying. Kumar et al. [86], classification and analysis of go-arounds in commercial aircraft using ADS-B data, effect detection in aviation composite structures using ML algorithms. Ai et al. [3, 4] use ML and natural language processing to anticipate human elements in aviation incident reports. Luo et al. [110] used event sequence-based technique for discovering the causality of the flight service process. Puranik et al. [143] use supervised ML to predict safety essential landing variables online in aviation. Truong and Choi [160] researched using ML methods to forecast the probability of tiny UAV intrusions into the National Airspace System. Musci et al. [121] studied ice detection on aircraft surfaces using ML algorithms based on hyperspectral and multispectral pictures. Li et al. [91] applied a logarithmic multivariate Gaussian model to analyze operational and mechanical abnormalities in commercial scheduled flights. Wu et al. [176] researched using a deep compression autoencoder network to recognize pilot tiredness levels. Kearney et al. [74] studied how alert designs affect air traffic controllers' eye movement patterns and situational awareness. Kuhn's [84] research used structured topic modeling to uncover latent subjects and patterns in aviation incident reports. Binias et al. [14] used ML strategy for recognizing airplane pilots' reactions to unexpected occurrences using EEG data.
 - **Aviation Operational Efficiency:** Piao et al. [141] used smooth motion planning learning for innovative air transport vehicles using Stable Auxiliary Gradient. Odisho et al. [129] studied the application of ML to improve runway safety by reducing runaway risk. Smirnov [155] studied the use of supercomputers and AI to ensure the safety of space flights.
 - **Safety and Risk Management:** Bi et al. [13] proposed a method based on IPPG that detects fatigue using driver face changes. Shafi et al. [154] used an artificial neural network model for spare parts prediction and classification, which impacted aviation safety. Blanco et al. [15] conducted a study on measuring cognitive workload using passive, dry EEG measurements in simulated flights. Rahmani and Weckman [146] investigated occupational safety risks for commercial drone pilots. Saleh et al. [152] studied data-driven insights and analysis on tail rotor failure in helicopter crashes.
 - **Technology and Automation in Aviation:** Holford [60] studied ethical investigation on the effects of cockpit automation on airline pilots' responsibilities. De Giorgi et al. [36] studied engine health monitoring using an ANN combined with kernel-based principal component analysis. Jiang et al. [70] developed a method to determine the intelligence level of pilots using visual gaze and operational behavior data.
- Unlike traditional ML methods, deep learning uses multi-layer artificial neural networks to identify and learn complex patterns on large amounts of data. This technique is also attracting increasing attention in the aviation industry. In particular, DL algorithms are of great importance in areas,

such as aviation safety, traffic management, pilot training, and research. DL has become a powerful tool for extracting information from complex data sets and transforming this information into valuable insights on aviation safety, efficiency, and security. In this context, the potential and applications of DL in the aviation industry remain an exciting area for future research and development. The classification helps understand the different methodologies and techniques applied in aviation safety and operations and highlights areas of focus for research and development in this area.

- **DL in Aviation Safety and Operations:** Zeng et al. [188] focus on a wirelessly enabled asynchronous federative Fourier neural network for turbulence prediction in urban air mobility, with an appropriate geographic model for wireless connection performance. Sui et al. [157] suggest using deep reinforcement learning to resolve conflicts in air traffic control, representing controllers' conflict resolution process as a Markov Decision Process. Khattak et al. [76–80] investigate turbulence intensity in the runway plan using a novel tree-structured Parzen estimator (TPE) optimized Kernel and Tree Boosting (KTBoost) model. Chiu and Lai [28] use deep learning techniques to detect and analyze unstable approaches in Taiwan's air transportation risk management. Kong et al. [87] assess airplane hard landing safety using Bayesian deep learning. Memarzadeh et al. [118] propose a semi-supervised deep learning model for detecting aircraft anomalies that are both reliable and explainable. Gao et al. [51] developed a deep learning-based wind field prediction approach emphasizing highly variable occurrences. Liu et al. [105, 106] describe a drone-based method for inspecting aircraft surface defects utilizing external restrictions and deep learning. Zhang et al. [193–196] use deep learning for aero-motor occlusion defect detection. Zhang et al. [197, 198] recognize aircraft skin interior state using deep learning. Lee et al. [90] utilize deep spatio-temporal neural networks for risk prediction and decision-making in aircraft operations. Dong et al. [43, 46] review and provide an outlook on DL in aircraft design, dynamics, and control. Ai et al. [3, 4] use a deep learning architecture with multi-sensor fusion data to recognize aircraft wake vortices. Dong et al. [43, 46] propose a deep learning strategy to identify reasons from accident records. Lin et al. [98] propose a DL-based safety monitoring framework for air traffic control systems. Yetgin et al. [185] apply DL to recognize electric lines in aerial pictures. Ni et al. [124] examine deep belief networks and principal component analysis in civil aviation safety assessment.
- **Bayesian and Probabilistic Models in Aviation:** Kong et al. [87] apply Bayesian deep learning to assess airplane hard landing safety. Li et al. [91] investigate using Bayes-

ian neural networks to predict flight routes and evaluate safety.

- **RL (Reinforcement Learning) in Air Traffic Control and Safety:** Sui et al. [157] examines deep reinforcement learning is proposed to resolve conflicts in air traffic control, modeling the conflict resolution process as a Markov Decision Process. Zhao et al. [201] investigated RIS-enabled air-to-ground NOMA communication with a distribution-robust DRL approach.
- **Hybrid and Ensemble Methods:** Su et al. [156] propose an improved gray correlation analysis and a combined prediction model for aviation accident prediction. Memarzadeh et al. [118] suggest a semi-supervised DL model for detecting aircraft anomalies. Lin et al. [98] introduce a DL-based safety monitoring framework for air traffic control systems. Chen et al. [21] investigated airplane track anomaly detection approach using deep mixed-density mesh networks. Dong et al. [43, 46] combined deep learning architecture and multi-sensor fusion data to recognize aircraft wake vortices. Ni et al. [124] examine deep belief networks and principal component analysis for civil aviation safety assessment.

3.3 Other Methods Used for Aviation Safety

Groupings have been made around specific themes to provide a clearer picture of origins, development, current status, and trends of aviation research. These groups will help understand the history, the current status, and the possible future developments of aviation safety-related research.

- **Origins and Historical Development of Research**

Early Technological and Safety Improvements

Aviation research in the early 2000s focused on enhancing safety and operational efficiency. These studies aimed to improve safety through technology-driven advancements. Gandhi et al. [50]: this study focused on obstacle detection using dynamic programming methods in aviation. It represents an early example of efforts to enhance safety through advanced algorithms. Oza et al. [133]: developing algorithms for classifying health and safety documents for aviation systems showcases the early application of data analytics and machine learning techniques.

Operational Efficiency and Safety

By the 2010s, research had increasingly focused on improving operational efficiency and solidifying safety measures. Kwong et al. [88]: using a chaotic oscillator and artificial neural network for wind shear warning demonstrates how artificial intelligence can be applied to aviation safety.

- Technological Innovations and Advanced Research

Applications of Artificial Intelligence and Machine Learning

Artificial intelligence, machine learning, and data science techniques have become more prevalent in aviation, particularly in safety, maintenance, and operational optimization. Ali [6]: the study on improving bagging ensemble performance using Bayes' rule contributed significantly to detecting abnormal movements, showing how machine learning methods can enhance flight safety. Deepa and Sudha [37]: this study on optimizing PID parameters for controlling aircraft dynamics was essential in ensuring aircraft stability, highlighting the modern application of classical control theory. Wu et al. [175]: the study on the sequencing model of an aircrew landing on an aircraft carrier significantly contributes to the planning and safety optimization of complex flight operations.

Human–Computer Interaction and Cognitive Systems

Human–computer interaction and cognitive systems became increasingly important during this period, especially in single-pilot operations and intelligent cockpit designs. Liu et al. [103]: the cognitive pilot–aircraft interface concept for single-pilot operations underlines the role of human–computer interaction in future aviation operations. Zhang et al. [193–196]: the study on evolutionary gaming and cooperative mechanisms in human–computer interaction offers valuable insights into the future development of intelligent cockpit designs. Naor et al. [123]: the case study on psychological safety within aviation new product development teams highlights how team dynamics and safety are crucial in innovative processes.

- Current Trends and Present Status

Data Science and Big Data Analytics

Data science and big data analytics are increasingly critical in optimizing aviation safety and operations. Rey et al. [147]: the data-driven methodology for detecting safety risks through aviation data analysis illustrates the growing use of big data analytics in aviation. Perboli et al. [139]: applying Natural Language Processing (NLP) techniques to detect human factors in aircraft accidents represents a crucial step toward minimizing human errors.

System Dynamics and Game Theory

System dynamics and game theory make aviation operations more complex and adaptive in this period. Patriarca et al. [136]: the study on ex-ante risk management demonstrates how system dynamics can be used to manage risks proactively. Xue et al. [179]: the

study on optimal allocation of take-off time in severe weather conditions emphasizes the importance of uncertainty management in operational planning.

- Future Trends and Evolution of the Research Field

Physics-Based Models and Deep Learning

In future, physics-based models and deep learning techniques are expected to find broader applications in aviation. Saha et al. [150]: estimating turbulent force using physics-based deep learning shows how machine learning and physics-based modeling are becoming critical in aviation applications.

Small Aircraft and Integrated Systems

Integrated systems for small aircraft are emerging as a critical research area aimed at enhancing the operational efficiency of these vehicles. Di Vito et al. [40]: the design advances in integrated mission management systems for small air transports indicate future trends in enhancing operational efficiency for these vehicles. Ma et al. [112]: examining emerging technologies to deal with uncertainties in aircraft maintenance routing highlights future research directions for optimizing maintenance processes.

The evolution of aviation research shows a progression from early efforts to enhance safety and operational efficiency to the application of emerging technologies like AI, data science, and human–computer interaction. Future advancements in these areas are expected to contribute significantly to operational optimization, safety improvements, and more effective management of human factors in the aviation industry.

4 Bibliometric Analysis of AI in Aviation Safety

Bibliometric analysis is used to identify publication trends, most frequently cited studies, and critical research teams in aviation safety. This analysis can reveal which topics are receiving more attention and which areas need more research [38, 39].

4.1 Performance Analysis for Aviation Safety and AI

The annual growth rate of publications, the journal with the highest number of authors and citations, and the institution and document with the highest number of publications and citations were identified. Authors and nations with the highest production of publications and citations were identified. The performance of the data was evaluated by keyword analysis.

4.1.1 Publication Trends

Figure 4 depicts the rise of documents over time in the bibliometric analysis of aviation safety-related research.

According to Fig. 4, scientists worldwide have a growing interest in analyses related to aviation safety. The number of publications related to aviation safety has seen a significant increase in recent years. While there were only 13 publications in 2020, this number rose to 68 in 2023. This growth, particularly in 2023, highlights how analyses related to aviation safety have increasingly incorporated advanced technological methods, such as AI, ML, and DL. The year 2023 stands out as the period with the most significant increase in research in this field. During this time, numerous publications explored how AI and ML techniques are utilized in aviation safety analyses.

4.1.2 Country or Regional Analysis

The productivity of countries for aviation safety studies is shown in Table 2.

Table 2 shows the countries broadcasting aviation safety. China is the most active Country in aviation safety, with 464 publications. The USA follows with 182 publications, Italy with 30, and India with 20. China has collaborated extensively with Hong Kong, the USA, Canada, and the United Kingdom in aviation safety studies, demonstrating a global effort to integrate AI and ML techniques into aviation safety.

In the cooperation map between countries, dark blue indicates countries that publish a lot, blue indicates countries that publish less, and gray indicates countries that do

Table 2 Country scientific production

Region	Freq	Region	Freq
China	464	Austria	5
USA	182	Brazil	5
Italy	30	Japan	5
India	20	Switzerland	5
Turkey	14	Finland	4
UK	13	Croatia	3
Portugal	12	Iceland	3
Canada	10	Pakistan	3
France	9	Saudi Arabia	3
Iran	9	Iraq	2
Israel	8	Singapore	2
South Korea	8	Angola	1
Germany	7	Colombia	1
Australia	6	Ireland	1
Belgium	6	Kenya	1
Malaysia	6	Netherlands	1
Poland	6	United Arab Emirates	1
Spain	6		

not. The thickness of the brown line used for cooperation between countries represents the strength of the relationship. The cooperation map of the countries that use the highest number of documents in cooperation is shown in Fig. 5.

Countries with great connectedness are those that cooperate with others the most. It is seen that China cooperates with Hong Kong, the USA, Canada, and the United Kingdom in studies related to aviation safety. The United

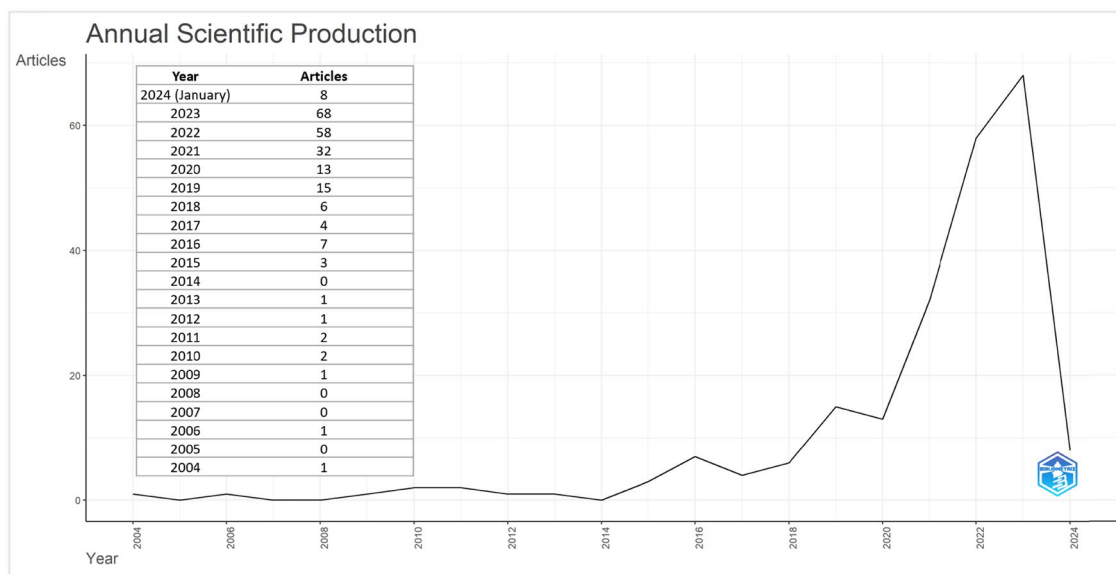


Fig. 4 Distribution of the studies used in the research according to years

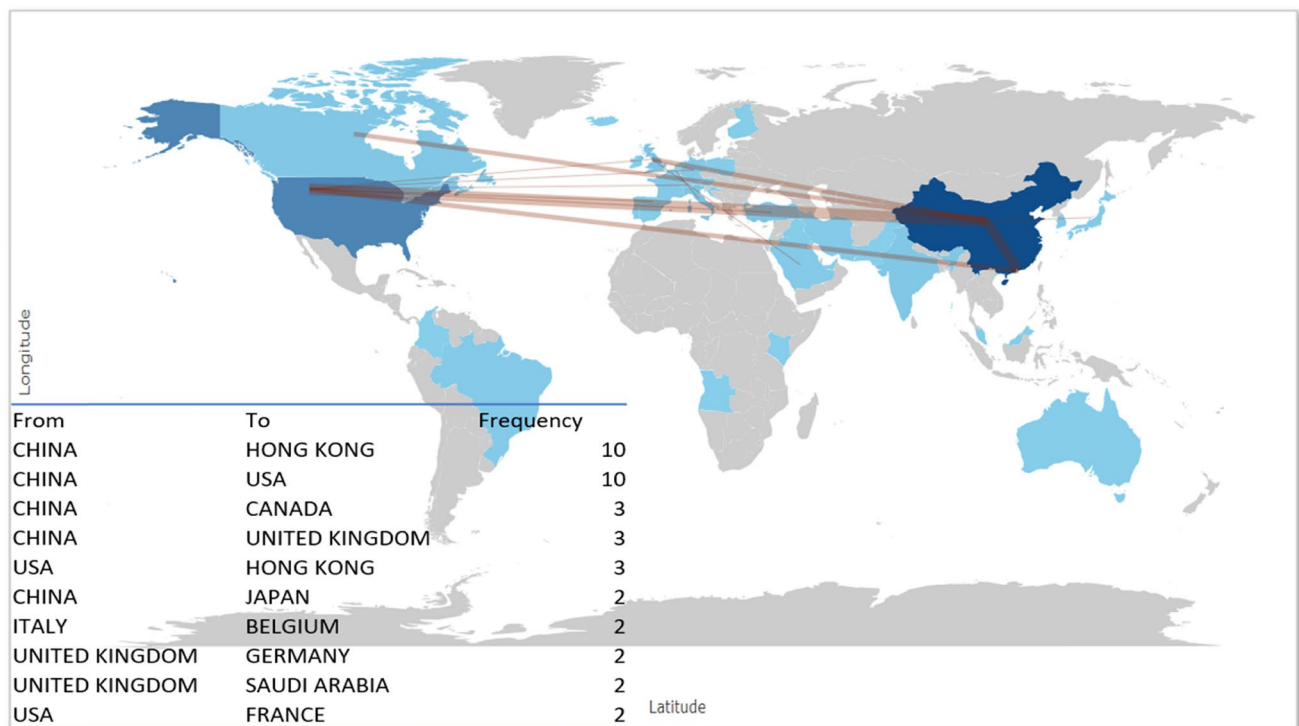


Fig. 5 Country collaboration map

Table 3 Related organizations that contributed to the research on aviation safety

Affiliation	Articles
Nanjing University of Aeronautics and Astronautics	50
Civil Aviation Flight University of China	35
Beihang University	22
Tongji University	19
National University of Defense Technology	16
Arizona State University	14
Hong Kong Observatory	11
Northwestern Polytechnical University	11
Southeast University	11
Vanderbilt University	11

Kingdom collaborates with Germany and Saudi Arabia, facilitating cross-border impacts of AI-based research. These collaborations indicate that the influence of AI and ML on aviation safety is being addressed globally.

4.1.3 Institutional Analysis

Table 3 shows the publishing outputs of the authors' institutions or related organizations that contributed to aviation safety research.

Table 4 The most relevant authors

Authors	Articles	Total citations
Wang H	11	81
Chan P-W	8	9
Zhang J	8	48
Chen F	7	7
Khattak A	7	7
Zhang Y	7	58
Li X	6	31
Liu X	6	70
Sun Y	6	20
Zhang X	6	241

Between 2004 and 2024, “Nanjing University of Aeronautics and Astronautics” ranks first with 50 publications, “Civil Aviation Flight University of China” ranks second with 35 publications, and “Beihang University” ranks third with 22 publications.

4.1.4 Author Analysis

A total of 224 research papers on aviation safety have appeared in different publications and were authored by 685 people. Table 4 shows the most relevant writers regarding citations and publications worldwide.

In terms of the number of publications, Wang H ranks first with 11 articles, Chan P-W and Zhang J rank second with eight articles, Chen F, Khattak A and Zhang Y rank third with seven articles. These authors are recognized for their work analyzing the impact of AI and ML on aviation safety.

Zhang X ranks first with 241 citations, Wang H ranks second with 81 citations, and Liu X ranks third with 70 citations. These authors have produced highly influential publications demonstrating the critical role of AI and ML techniques in aviation safety.

4.1.5 Journal Analysis

The ranking of publications on aviation safety according to the number of publications is given in Table 5.

According to Table 5, IEEE Access ranks first with 58 citations in aviation safety, Sensors follows with 50 citations, and Safety Science has 46 citations. The articles published in these journals significantly contribute to understanding how AI and ML techniques are utilized in aviation safety.

Regarding the number of articles published, Aerospace ranks first with ten articles, followed by Atmosphere and Sensors with 9 and 7 articles, respectively.

Table 5 Most published and cited journals for aviation safety

Sources	Number of publications	Total citation
<i>Aerospace</i>	10	40
<i>Atmosphere</i>	9	27
<i>Sensors</i>	7	50
<i>Applied Sciences (Switzerland)</i>	6	8
<i>IEEE Transactions on Intelligent Transportation Systems</i>	6	15
<i>Safety Science</i>	6	46
<i>Transportation Research Part C: Emerging Technologies</i>	6	17
<i>Engineering Applications of AI</i>	4	2
<i>IEEE Access</i>	4	58
<i>Reliability Engineering and System Safety</i>	4	7

Table 6 The most efficient documents

Paper	DOI	Total citations
Kuhn [84], <i>Transp Res Part C Emerg Technol</i>	10.1016/j.trc.2017.12.018	116
Zhang [193], <i>Decis Support System</i>	10.1016/j.dss.2018.10.009	109
Balasingam [12], <i>Int J Clin Pract</i>	10.1111/ijcp.12989	96
Zhang [192], <i>Decis Support System</i>	10.1016/j.dss.2020.113246	90
Liu [105], <i>Knowl Based Syst</i>	10.1016/j.knosys.2016.08.031	73
Groth [53], <i>Reliab Eng Syst Saf</i>	10.1016/j.res.2010.06.005	71
Ma [112], <i>Complexity</i>	10.1155/2018/3813029	66
Wang [173], <i>Comput Electron Agric</i>	10.1016/j.compag.2020.105523	63
Memarzadeh [117], <i>Aerosp</i>	10.3390/AEROSPACE7080115	56
Wu [176], <i>IEEE Trans Instrum Meas</i>	10.1109/TIM.2018.2885608	43

4.1.6 Analysis of References

The most productive article is the one with the most significant total citations. Table 6 shows the top ten most productive articles.

The article “Using Structural Topic Modeling to Identify Latent Topics and Trends in Aviation Incident Reports” by Kuhn [84], published in *Transportation Research Part C: Emerging Technologies* ranks first with 116 citations. This article follows “Ensemble ML Models for Aviation Incident Risk Prediction”, published in *Decision Support Systems* by Zhang and Mahadevan [190] with 109 citations.

4.1.7 Keyword Analysis

A keyword describes the article as a word or phrase. The existence of a keyword in an article determines how widely a phrase is used. VOSviewer was used to determine the combined impact of working with keywords and the research knowledge base. Based on the indexed keywords of the article, 740 keywords were discovered. Figure 6 shows 41 keywords evaluated for analysis after setting the threshold to 3 in the VOSviewer program.

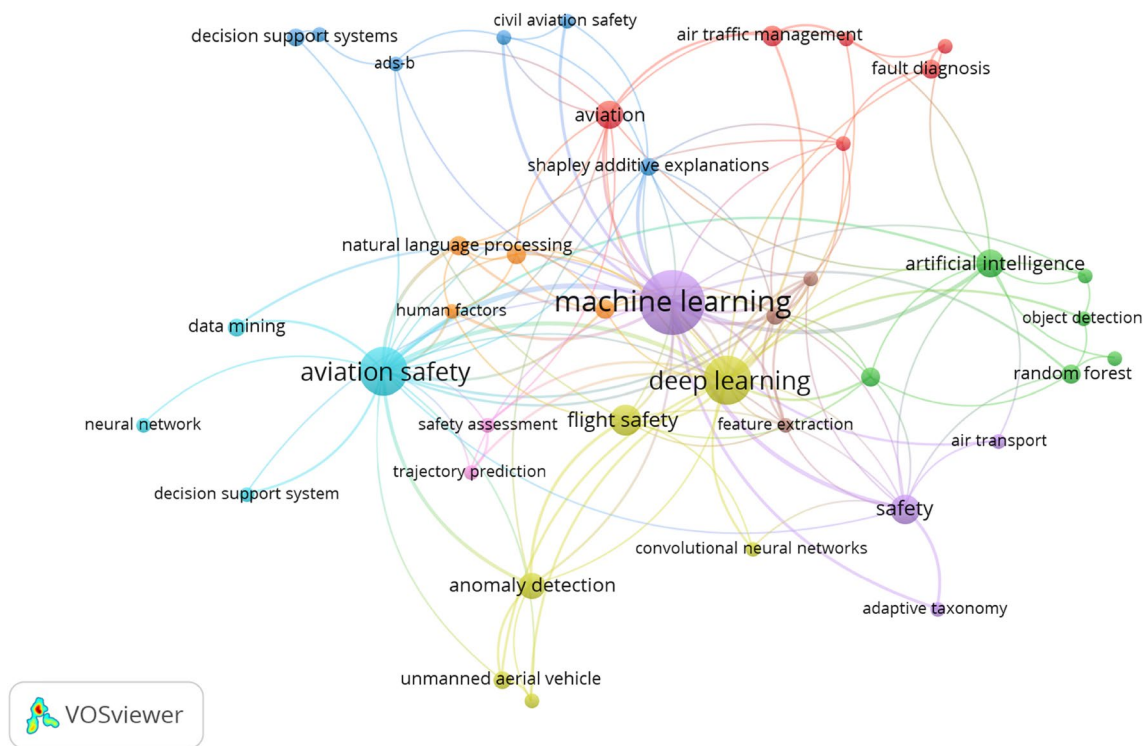


Fig. 6 Co-occurrence keywords

A different set of keywords is created with each color. In Fig. 4, each circle indicates the presence of a particular keyword and sub-domain of the aviation safety network theme. A circle with a similar color represents the distribution in a comparison region. The largest cluster is named “machine learning” according to the number of elements. This cluster contains keywords, such as “adaptive taxonomy, air transport, and safety”. This highlights the critical role of ML in aviation safety. The second largest cluster is called “aviation safety”. It contains keywords like “data mining, neural network, decision support system”. This illustrates how AI and ML techniques enhance efficiency and minimize risks in aviation safety. The third-largest cluster is named “deep learning”. It contains keywords, such as “unmanned aerial vehicle, aviation safety, and anomaly detection”. The third-largest cluster is named “deep learning”, with keywords like “unmanned aerial vehicle, aviation safety, anomaly detection”. This shows that DL is crucial, particularly in UAVs and anomaly detection areas.

With the VOSviewer software, the keywords “Overlay visualization” were colored differently according to the year of publication, and the time intervals in which they appeared in the literature were determined. In our case, the average (yellow) publication year for newly introduced terms is 2023. The colors of the items were determined by the time elapsed since their publication. The period from 2018 to 2023 (Blue–Green–Yellow Color) is shown in Fig. 7.

While previously used keywords, such as “aviation safety, prediction, and decision support system”, are keywords that have been intensively studied in the 2018–2020 period, it can be said that keywords, such as “unmanned aerial vehicle, eeg, mental states classification” have also been used in the literature recently.

4.2 Scientific Mapping Analysis for Aviation Safety and AI

Scientific mapping is the use of computer tools in general to display, analyze, and simulate diverse scientific and technological processes. This method helps researchers to see relationships, patterns, and trends more clearly across large data sets. Scientific mapping often visually reveals relationships between texts, similarities or differences between topics, publication trends, and the importance of key concepts. Scientific mapping is a critical tool for determining complex relationships, data-driven analyses, and priorities in decision-making processes [39].

4.2.1 Three Field Plots

“Three field plots” are often a preferred visualization method to show relationships between topics, keywords, or articles during a literature review. This is used to compare or analyze data representing different research areas

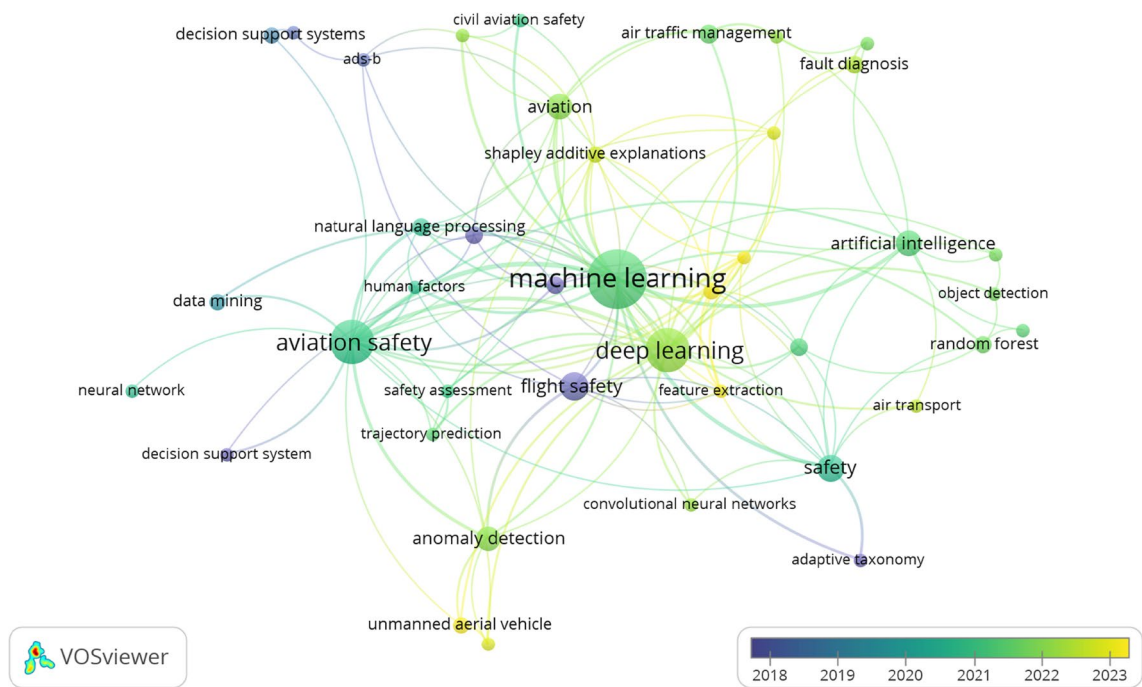


Fig. 7 Keyword timeline

visually. The three parameters to be related in this so-called “Sankey diagram” (keyword, author name, and Country) are specified in the software, and the leading ones for each parameter are shown in Fig. 8.

The size of the boxes in Fig. 8 represents the magnitude of the link between the parameters. In this graphic, the size of the boxes represents influential factors in the literature. The leading Country is “China”, the lead author is “Wang H” and the most used keyword is “machine learning”.

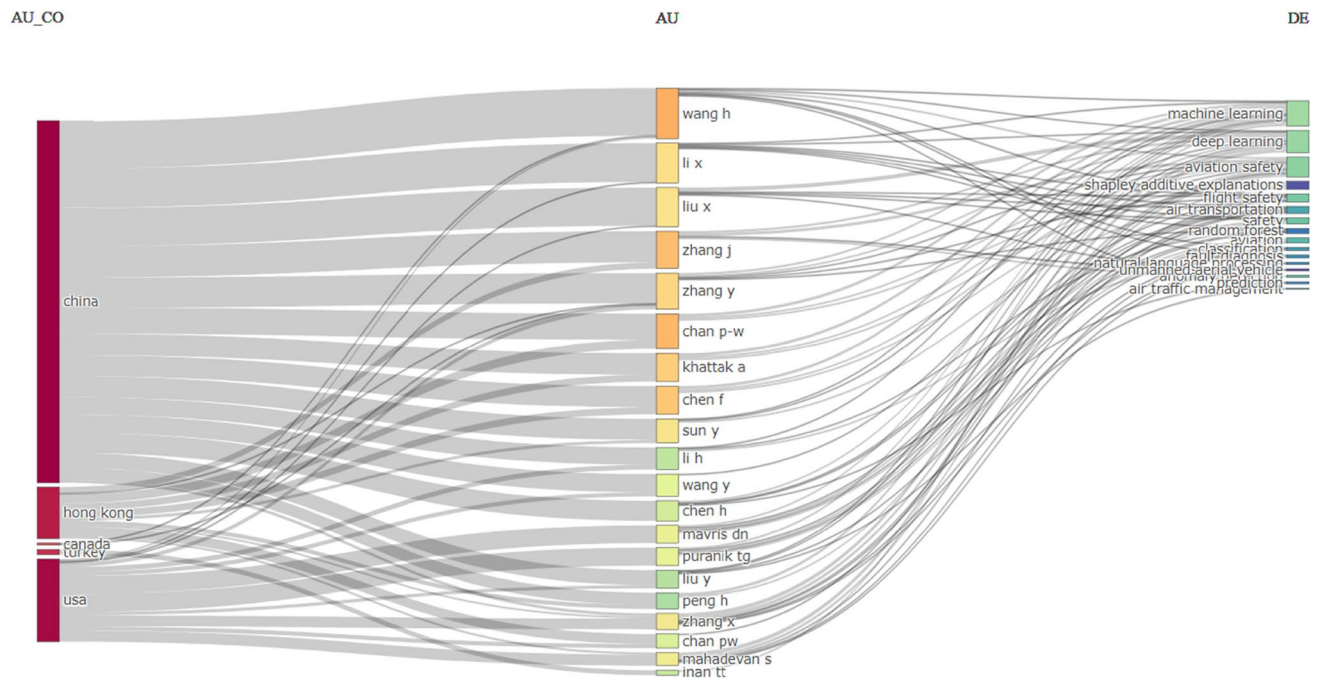


Fig. 8 Three field plots

4.2.2 Factor Analysis

Factor analysis is a statistical technique within such tools and is often used to reduce complexity and reveal hidden structures in multivariate data sets. It is a method used to understand the relationships between variables in a data set and to make the data set more understandable and manageable. This analysis is usually expressed in coefficients called factor loadings. These loadings indicate which factors the variables are related to or which factors explain the commonality between certain variables. The factor analysis of the keywords is given in Fig. 9.

As a result of the factor analysis, when the keywords of the articles on aviation safety are analyzed, it is seen that some concepts are in the first cluster with high factor loads; keywords, such as “machine learning, deep learning, and transfer learning”, are collected in the red cluster. The keywords “pilot deficiencies, eeg, mental states classification” are in the blue cluster. The keywords “behavioral sciences, time series analysis, unmanned aerial vehicle” are in the green cluster.

4.2.3 Thematic Map

A “thematic map” is a graphic created to visually represent the relationships between words, topics, or keywords used in the research literature. In order to identify the main research topics of the field, a thematic review of articles related to aviation safety was conducted using the author’s keywords. Figure 10 shows the related studies are grouped under four themes with different intensities.

Considering the author’s keywords, keywords, such as “air traffic management, aviation safety, fault diagnosis,

machine learning, and deep learning”, are at the forefront of scientific studies.

4.2.4 Trend Topics

The trending topics of aviation safety in the literature by years from 2004 to 2024 are given in Fig. 11.

The top trending topic in the first quarter of 2021, the second quarter of 2022, and the third quarter of 2023 was “ML”. The keyword “DL” was the trending topic in the first quarter of 2022, the second quarter of 2023, and the third quarter of 2023. Currently, “ML, DL, aircraft, aviation safety, long short-term memory” is the most trending topic in this area.

4.2.5 TreeMap

A TreeMap is a type of chart used to visualize hierarchical data structures. Data are represented using areas of rectangles. These rectangles represent categories or subcategories in the hierarchical structure. Large rectangles usually represent broader categories, while small rectangles represent subcategories or subdivisions. Colors and size of the rectangles can be used to indicate the value or the size of each category or subcategory. The TreeMap’s frequency of use of keywords is given in Fig. 12.

The blue rectangle in Fig. 12 shows that “machine learning” has the highest use rate of 8% (corresponding to 8% of the total keywords). The keyword “deep learning” has a use rate of 7%. The keyword “aircraft” has a use rate of 5%. The keywords “aviation safety” and “artificial intelligence” have a use rate of 4%.

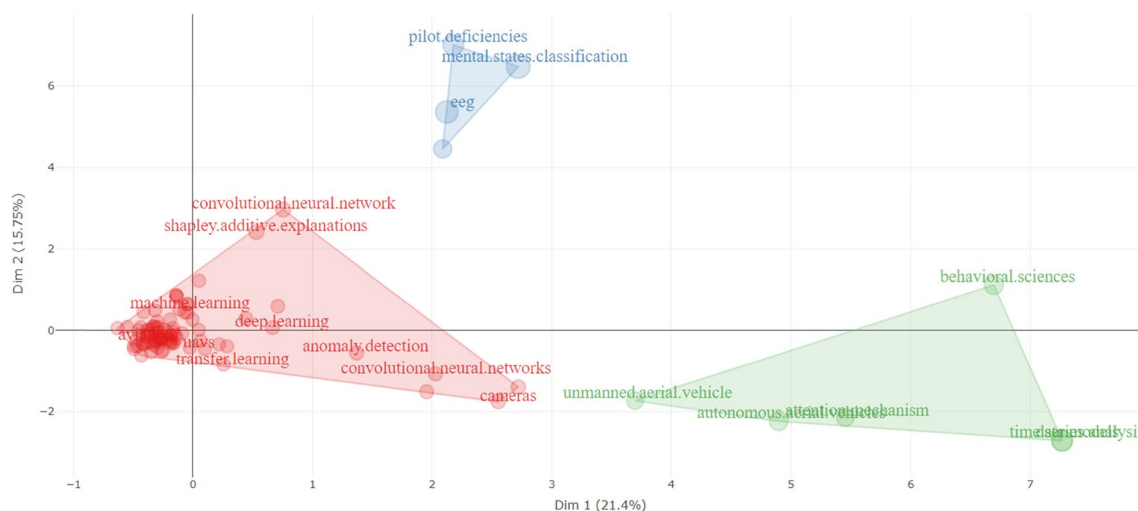


Fig. 9 Factor analysis of author keywords

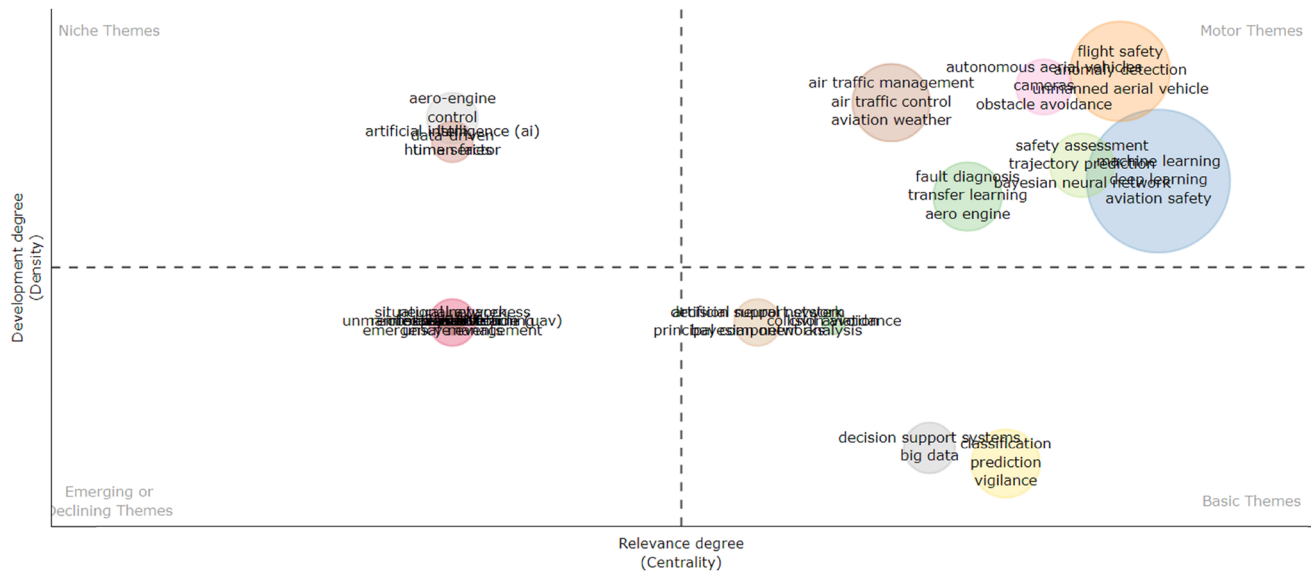


Fig. 10 Thematic map based on keywords

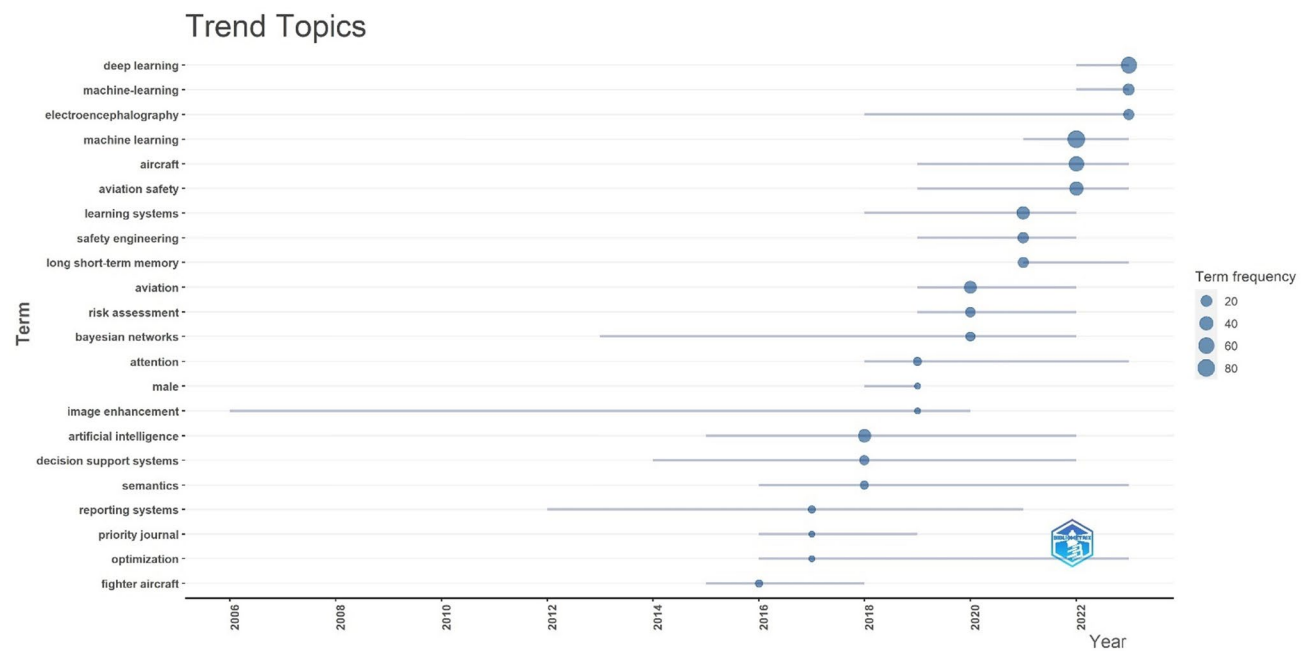


Fig. 11 Trending topics by year

5 Discussion

This study includes a systemic review and bibliometric analysis of the literature on aviation safety was published between 2004 and 2024. This article focuses on publications indexed in the Scopus database on AI, ML, and DL used in aviation safety studies. The literature review

during this period was evaluated by bibliometric analysis of scientific productivity, keywords, publication trends, and vital factors related to aviation safety. With systematic analysis, the focal points of research, techniques, and trends were examined. The increase in scientific output on issues related to aviation safety is occurring at a remarkable pace. There are few studies analyzing bibliometric data of research articles in different fields. In this regard,

these researches focus on, the leading countries and institutions, the keywords concentrated on in the literature, and the journals and conferences where these publications interact.

5.1 Identified Trends and Gaps in the Literature

The identified trends and gaps in the literature are based on current research intensities and the developments in these areas over time.

Identified Trends:

- **AI and ML:**
Trend: In recent years, ML and DL techniques have become some of the most frequently used methods in aviation safety research. This area is continually evolving and offering innovative approaches.
Gap: More research is needed on AI and ML models' real-time applicability and reliability.
- **Aviation Safety:**
Trend: Aviation safety is a critical research area for the aviation sector. Research focuses on accident analysis, pilot behavior, and improving safety standards.
Gap: There is a need for more advanced analysis of pilot behavior and simulation techniques.
- **Technological Methods:**
Trend: Advanced technological methods, such as electroencephalography and long-short-term memory models, are used in aviation safety research.
Gap: More empirical data are needed on the effectiveness and applicability of these technologies.
- **Decision Support Systems:**
Trend: Decision support systems and risk assessment methods support decision-making processes in the aviation sector.
Gap: Further research is required on integrating decision support systems and making them more user-friendly.

5.2 Future Research Agenda

A future research agenda has been proposed based on the identified trends and gaps.

- **Real-Time Artificial Intelligence Applications:**
RQ: How can artificial intelligence and machine learning models be more reliable for real-time aviation safety applications?
Method: Research should focus on real-time data analysis, simulation studies, and model optimization.
- **Pilot Behavior Analysis and Simulations:**
RQ: How can the simulation and analysis of pilot behavior contribute to improving safety standards?

Method: In-depth studies using advanced simulation techniques, neurophysiological modeling, and behavior analysis should be conducted.

- **Applicability of Advanced Technological Methods:**
RQ: How can electroencephalography and long-short-term memory models be used more effectively in aviation safety?
Method: Empirical research should be conducted on these technologies' practical applications and effectiveness.
- **Integrated Decision Support Systems:**
RQ: How can decision support systems be more user-friendly and open to integration?
Method: Studies should focus on user experiences, system design, and integration processes.

5.3 Enhancing Aviation Safety for Future Studies

For future studies, technical details, such as algorithm selection, training process, and model evaluation, in the use of ML and DL technologies can be discussed as follows:

- **Algorithm Selection**
 - **Problem Definition and Data Type:** *Algorithm Selection:* The choice of algorithm should depend on the problem definition and the type of data used. For classification problems, algorithms like SVM (Support et al.), Random Forest, or Logistic Regression can be preferred. CNNs (Convolutional et al.) or RNNs (Recurrent Neural Networks) may be used for tasks such as image or natural language processing. *Customized Models:* For specific problems, customized deep learning architectures (e.g., Faster R-CNN for aircraft surface defect detection) may be chosen.
 - **Model Complexity and Computational Power:** *Simple Algorithms:* Simpler and faster algorithms (e.g., k-Nearest Neighbors (k-NN) or Decision Trees) can be used for datasets with low volume and complexity. *Complex Algorithms:* For large and complex datasets, more sophisticated models, such as DNNs (Deep et al.), Transformers, or GNNs (Graph et al.), may be employed.
- **Training Process**
 - **Data Preprocessing:** *Data Cleaning:* It is crucial to clean the dataset before training, fill in missing values, and standardize data types. *Feature Engineering:* Engineering features from the data can significantly enhance algorithm performance. For example, time series features may be extracted for aircraft sensor data.

- **Model Training:** The training process starts with splitting the dataset into training, validation, and test sets. These steps are critical for measuring model performance and preventing overfitting. *Hyperparameter Optimization:* Techniques like Grid Search or Bayesian Optimization can be used to optimize hyperparameters and maximize algorithm performance. *Regularization:* Techniques such as Drop-out or L2 Regularization can be employed to prevent overfitting.
- **Data Augmentation:** *Data Augmentation Techniques:* Data augmentation techniques, especially in deep learning models, can improve model generalization when working with limited datasets. Techniques, such as transformations, cropping, rotation for image data, word replacement, and resampling, for text data can be applied.
- **Model Evaluation**
 - **Performance Metrics:** *Classification Problems:* Accuracy, F1 Score, ROC-AUC, Precision, and Recall can be used. *Regression Problems:* MSE (Metrics like Mean Squared Error), MAE (Mean et al.), and R-squared should be used to evaluate model performance. *Visual Data:* For image processing applications, metrics, such as IoU (Intersection over Union) and MAP (Mean et al.), can be utilized.
 - **Model Validation:** *Cross-Validation:* K-fold cross-validation can be used to test how the model performs on different subsets of the data, enhancing the model's generalization ability. *Test Set Performance:* The model's performance on the test dataset should be measured and compared with the results from the training and validation sets.
 - **Model Explainability and Reliability** *Model Explainability:* The model's decisions must be explained in critical areas like aviation. Techniques such as LIME or SHAP can be used to explain the model's decisions. *Safety and Reliability:* When used in safety-critical systems, the model should undergo robustness and stress testing to assess its performance in extreme conditions.
 - **Conclusions and Future Work:** *Model Development and Improvement:* Future work may focus on new data collection methods, using more sophisticated algorithms, and improving existing models. *Transfer Learning and Adaptation:* Research may explore adapting pre-trained models to different problem areas (Transfer Learning). *Real-Time Applications:* For areas such as air traffic control or flight safety, efforts could be made to optimize real-time data processing and decision-making time.

These steps will guide effectively utilizing ML and DL technologies in safety-critical domains such as aviation.

5.4 Study Limitations

This study provides a comprehensive systematic review and bibliometric analysis of aviation safety literature, focusing on AI, ML, and DL techniques. However, several limitations should be acknowledged:

Database Limitation: The study primarily relied on the Scopus database for data collection. While Scopus is a comprehensive and widely used database, it may only cover some relevant publications. Other databases, such as Web of Science, Dimensions, Cochrane Library, and PubMed, although considered in the context of future research, were not included in the current analysis. This could result in omitting relevant studies that might be indexed in these databases but not in Scopus.

Language Restriction: The study only included articles published in English. As a result, relevant research published in other languages may have been excluded, potentially limiting the global perspective on aviation safety and the use of AI, ML, and DL techniques.

Time Frame Limitation: The literature review focused on 2004 and January 2024 publications. Although extensive, this period may exclude early foundational research or recent developments post-January 2024, leading to a gap in capturing the most current trends or historical context that influence present-day research.

Exclusion of Certain Document Types: The analysis was limited to peer-reviewed journal articles, excluding conference papers, theses, editorials, and other documents. Conference papers often introduce cutting-edge research and developments that might not yet be published in journals, thus potentially overlooking emerging trends and innovations.

Technological Bias: The emphasis on AI, ML, and DL techniques may have led to underrepresenting other significant methodologies and tools used in aviation safety research. This focus may create a technological bias, potentially overlooking valuable insights from studies employing traditional or alternative approaches.

Bibliometric Analysis Tools: The study utilized specific bibliometric analysis tools, including RStudio (Biblioshiny) and VOSviewer. While these tools are powerful and widely used, the choice of software may influence the interpretation and visualization of data. Different tools or methodologies yield slightly different insights or visual representations.

Lack of Qualitative Insights: While the study provided a quantitative bibliometric analysis, it lacked a qualitative assessment of the content, such as the included studies'

theoretical frameworks or methodological rigor. This limitation may reduce the depth of understanding regarding the quality and impact of the research.

Generalizability: The study's findings are specific to aviation safety research and may not be directly generalizable to other domains, even within the broader aviation field. This limits the applicability of the results to other areas of research or industry practices.

Dynamic Nature of the Field: The rapidly evolving nature of AI, ML, and DL in aviation safety means that the research trends identified in this study could quickly become outdated. This dynamic field requires continuous updates and reviews to stay relevant and accurate.

In summary, while this study offers valuable insights into the state of research on aviation safety, particularly regarding AI, ML, and DL techniques, these limitations highlight the need for ongoing research and a broader scope in future studies. Expanding the databases, including more document types, and incorporating qualitative assessments could provide a more comprehensive understanding of the field.

6 Conclusions

For systematic review results, the focal points of the research can be grouped under the following headings: aviation safety and data analysis, risk factors and analysis, pilot behaviors and safety, technology and design: classification and analysis methods, performance analysis and prediction, aviation safety and precautions, uncrewed aerial vehicles (UAVs) and security, operational processes and techniques: computing and data analysis. These cover various aspects and focus on broad research in aviation security and data analytics.

The techniques used in aviation safety research are diverse and are generally used in subjects, such as analyzing aviation data, improving aviation safety, and evaluating pilot behavior. These techniques include ML algorithms, time series models, deep learning techniques, AI techniques, neurophysiological modeling, optimization algorithms, text mining algorithms, and physical reverse learning machines. These techniques are critical in improving data analysis, security strategies, and operational decisions in the aviation industry. These techniques are essential in the aviation industry, requiring high precision and accuracy.

The aviation industry has continually focused on research and innovation regarding safety, efficiency, and technological advances. In this regard, recent research trends and academic studies have focused on critical issues, such as aviation safety, pilot behavior, and analysis and prevention of accidents. Prominent research trends include determining various contributing factors in aviation accidents, developing severity prediction models, examining the effectiveness

of deep learning-based time series models for analyzing and predicting accidents, and conducting more comprehensive research on pilot behavior and improving model performance. Development of AI models for analysis and classification of aviation accidents, analyzing the cognitive workload of pilots with EEG data and ML methods and contributing to the development of aviation systems, aviation safety, analysis of EEG signals, AI-supported security systems, analysis of aviation accidents, model explainability, causes of hard landings, air traffic management, complex system models, deep learning applications, monitoring pilots' emotional states and effectiveness of aviation training, icing modeling, ML algorithms, and aviation safety. These studies significantly contribute to increasing safety standards in the aviation industry, improving operational efficiency and the sector's future.

By summarizing in-depth research on aviation safety, we hope the findings will guide the rapidly expanding aviation industry for additional research directions and perspectives. Managerial implications for future studies are as follows: this study contributes valuable data to evaluating aviation safety in terms of the most influential sources, authors, linkages, nations, and studies in the available literature. This provides researchers and practitioners with guidance on which articles they should reference, which are most relevant, and which have the most impact on aviation safety. A bibliometric review of aviation safety in aviation studies can provide a comprehensive overview of past and current research and identify future research directions for aviation safety. This can guide researchers in understanding and further contribute to existing knowledge in this field. The findings of the study reflect the state of research on aviation safety.

As a reference point, this study provides researchers with a comprehensive understanding of AI, ML, and DL. This allows researchers to delve deeper into the topic. Using analysis of quotations and co-citations, researchers can identify different streams or areas of research that make up their intellectual landscape, allowing them to identify themes and insights. This can guide how researchers can relate their work to existing literature. Depending on the research area, it is possible to identify gaps in the literature and potential research directions. This provides professionals, educators, and students with up-to-date information on important AI, ML, and deep learning techniques they can apply in their future work within aviation safety. This research can give significant information to academics and practitioners studying aviation safety and indicate areas of research that deserve more attention to provide theoretical and practical consequences.

While the main focus of this study is the Scopus database, other sources, such as Dimensions, Web of Science, Cochrane Library, and PubMed, were also considered. The next potential step is to design and build graphical tools that

present more data and cover a broader area. This will expand the scope of the research and provide a more comprehensive assessment.

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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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